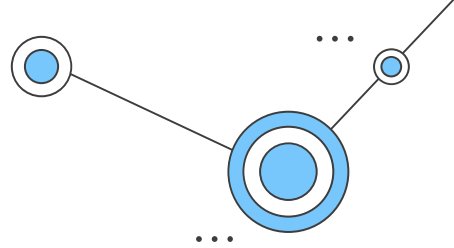


# Storage account



# Storage accounts

Cloud solution for storing data



Account in which have access to different types of storage services:

- Blobs (Containers)
- File Shares (File shares)
- Queues (Queues)
- Tables (Tables)

## Data storage

 Containers

 File shares

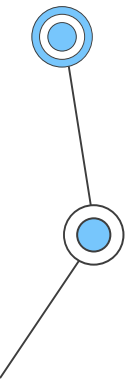
 Queues

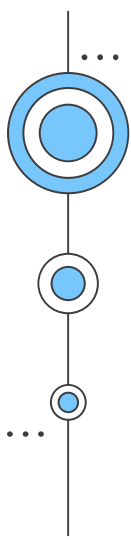
 Tables

⇒ There are other storage-related services (e.g. SQL databases)

⇒ Most important data storage service

- Access Tier (Hot, cool, archive)
- Redundancy options
- Disc Storage, File Sync, Data Transfer, Data Migration
- Database services



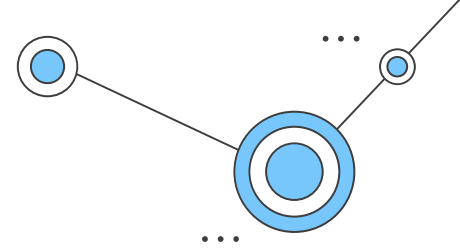


# Redundancy Options



# Redundancy Options

Protect from planned and unplanned events



Microsoft Azure

Search resources, services, and docs (G+/)

Home > Storage accounts >

## Create a storage account

Basics | Advanced | Networking | Data protection | Encryption | Tags | Review

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

### Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription \*

Resource group \*  [Create new](#)

### Instance details

If you need to create a legacy storage account:

Storage account name

Region

Performance

Redundancy

☒ Make read access to data available in the event of regional unavailability.

[Review](#) [< Previous](#) [Next : Advanced >](#)

Locally-redundant storage (LRS):  
Lowest-cost option with basic protection against server rack and drive failures. Recommended for non-critical scenarios.

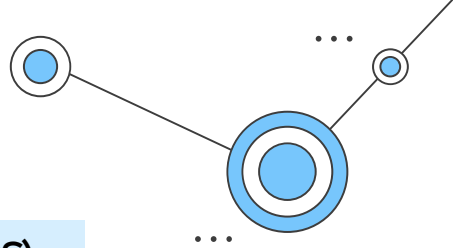
**Geo-redundant storage (GRS):**  
Intermediate option with failover capabilities in a secondary region. Recommended for backup scenarios.

Zone-redundant storage (ZRS):  
Intermediate option with protection against datacenter-level failures. Recommended for high availability scenarios.

Geo-zone-redundant storage (GZRS):  
Optimal data protection solution that includes the offerings of both GRS and ZRS. Recommended for critical data scenarios.

- Azure Storage always keeps multiple copies
- Trade-off:  
Higher availability + durability vs. cost

# Redundancy Options



## Locally redundant storage (LRS)

Three copies **within a single data center**

Lowest cost & least durability

Protects data against server rack and drive failures

Disaster in the data center: Data may be lost

## Zone-redundant storage (ZRS)

Three replications **across three Availability Zones**

Separate physical location

Protects data against disaster in a data center

## Geo-redundant storage (GRS)

Three copies using LRS in one region

Three copies using LRS in **secondary region**

Paired region based on region pairs

Read-access per default only after failure (RA-)GRS

Protects from regional disaster

## Geo-zone-redundant storage (GZRS)

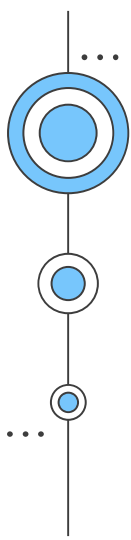
Three replications **across three Availability Zones (ZRS)**

Three copies using ZRS in **secondary region**

Maximum durability, availability and consistency

Read-access per default only after failure (RA-)GZRS



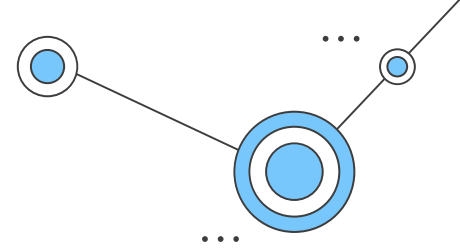


# Blob Storage



# Blob storage

(**B**inary **L**arge **OB**ject)



Solution to store massive amounts of unstructured data

⇒ Can be any type of data: Images, documents, backups, videos

Data storage

Containers

File shares

Queues

Tables

Containers are used to organize the files (like a folder)

Not hierarchical

**Storage Account**

**Container 1**

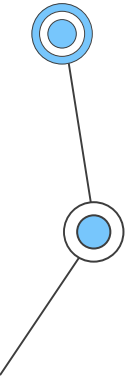
**File 1**

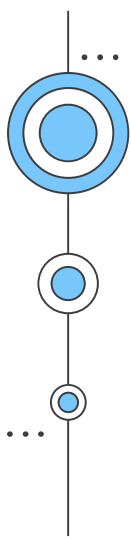
**File 2**

**File 3**

**Container 2**

**File 3**



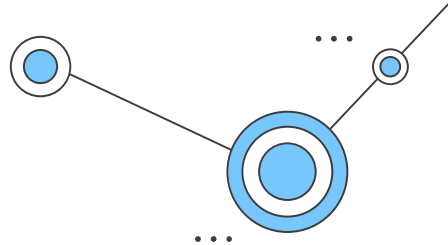


# Access Tiers





# Access Tiers



## Hot

More expensive storage cost  
Cheaper read/write operations  
Low latency  
Good for **frequently accessed** data  
Example: Images on a website

## Cool

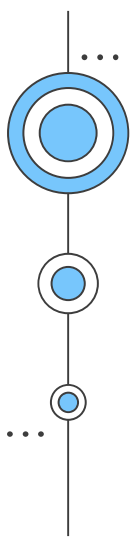
Cheaper storage cost than "Hot"  
More expensive read/write operations  
Higher latency  
Good for **infrequently accessed** data  
Example: Short-term backup  
Older datasets

## Archive

Cheapest storage  
Most expensive access cost  
Cannot be read directly (offline tier)  
Must be rehydrated to cool or hot  
before it can be accessed  
Example: **Long-term backup**  
Data Archiving

Default at account level: Hot or Cool



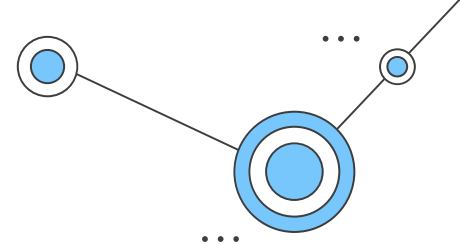


# Queue storage



# Queue storage

*storing large numbers of messages*



Create a backlog of work (messages)

Queue

Message 1

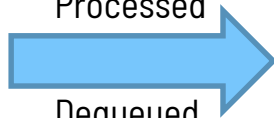
Message 2

Message 3

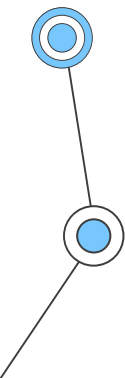
Queue 2

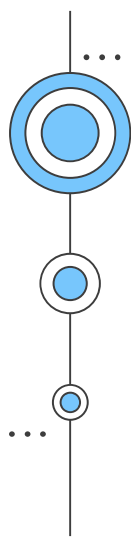
Message A

Processed



Dequeued



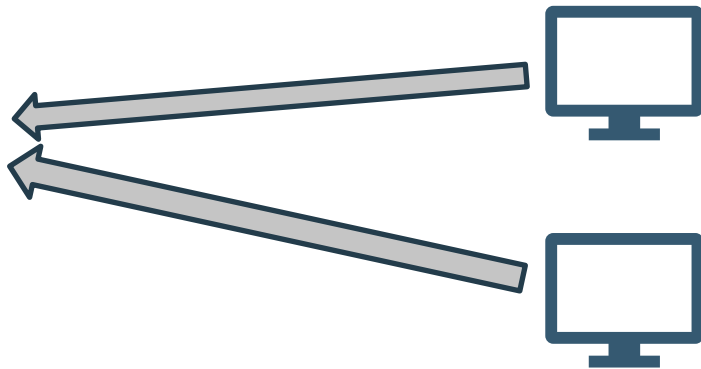
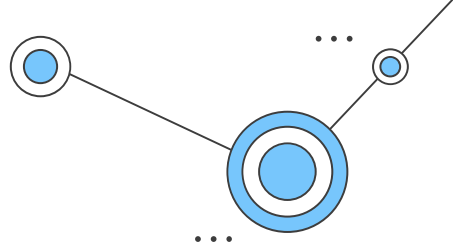


# Azure Files



# Azure Files

*Managed file shares in the cloud*

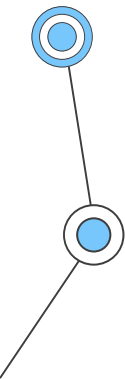


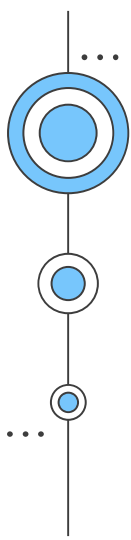
Can be mounted by cloud or on-premise

Replace or supplement on-premises file servers:



Can be accessed via *Server Message Block (SMB)* protocol  
or *Network File System (NFS)* protocol



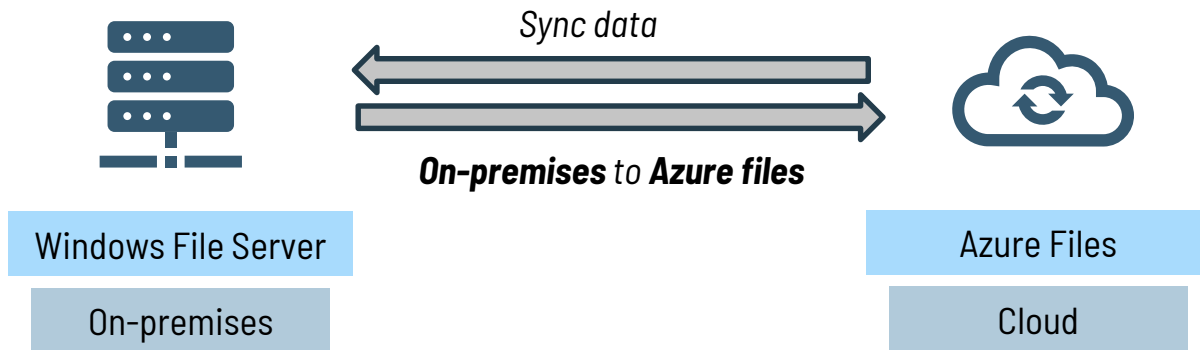


# File Sync



# File Sync

*Sync data from on-premises to Azure Files*

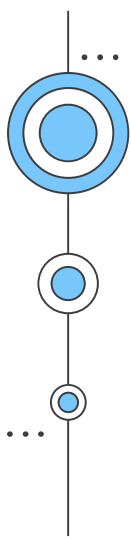


## Use-cases:

- Sync data across multiple sites/offices
- Disaster recovery

## How:

- Install File Sync agent on Windows file server
- Add it to Azure File Sync Deployment



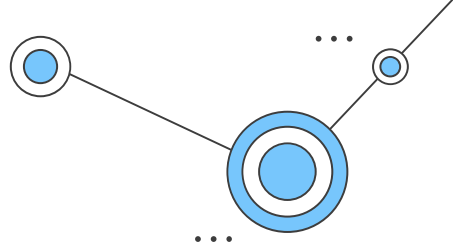
# Azure Tables





# Azure Tables

*NoSQL (non-relational SQL) database solution*



- Very inexpensive (NoSQL) database service
- Key/attribute data storage without schema
- Designed for high volumes of data

Redundancy options

High availability

## Use cases:

- Store large amount of structured data
- No need for complex joins
- [Alternative to Cosmos DB](#)

Storage Account



Tables



Tables



Tables

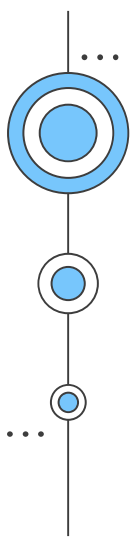
Rows = Entities

Attribute

Attribute

Key

| employee_id | first_name      | last_name        |
|-------------|-----------------|------------------|
| 1           | Attribute Frank | Attribute Miller |

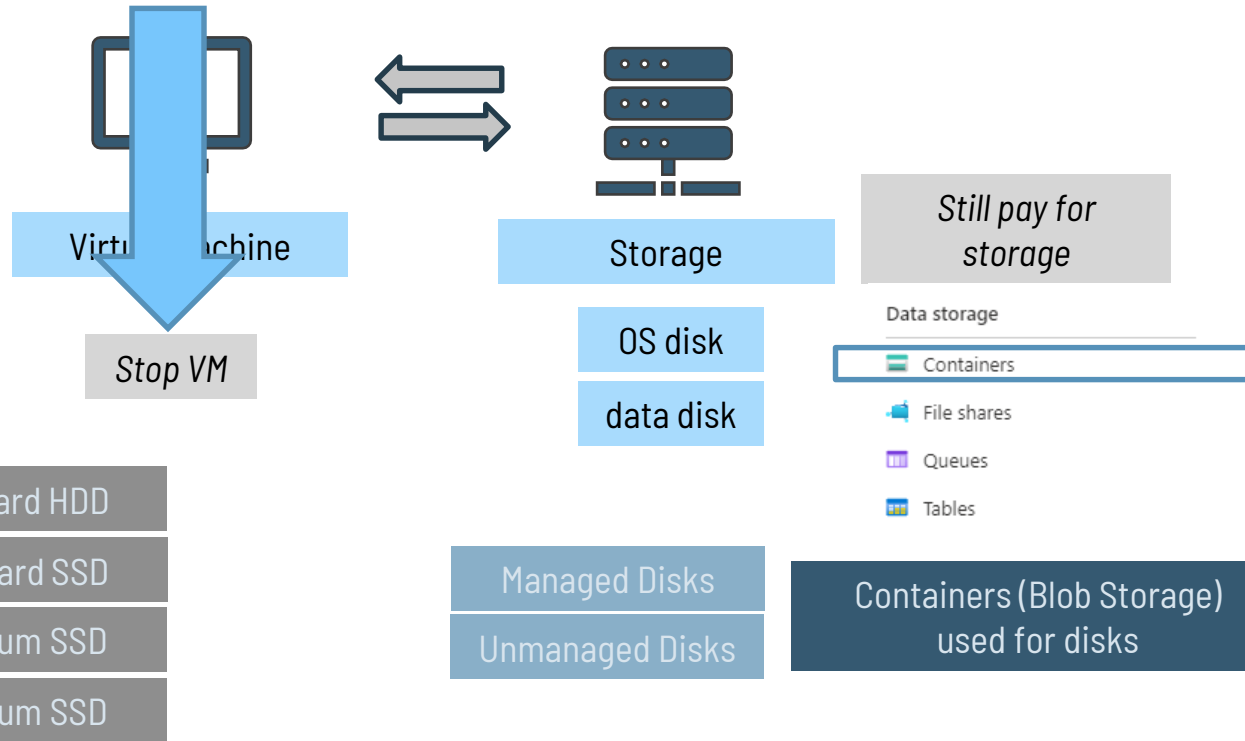
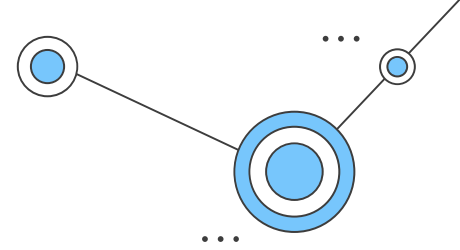


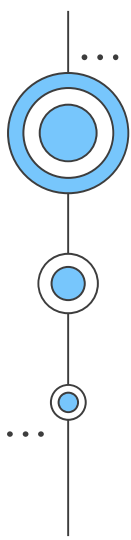
# Disk storage



# Disk storage

*Storage for virtual machines*



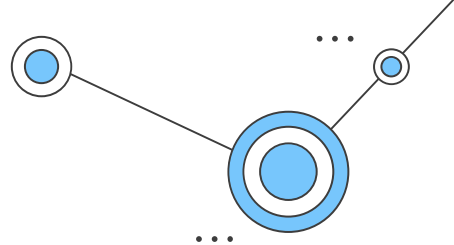


**AzCopy**



# AzCopy

*Command-line tool to copy data to and from storage accounts*

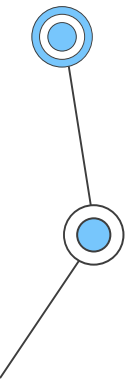


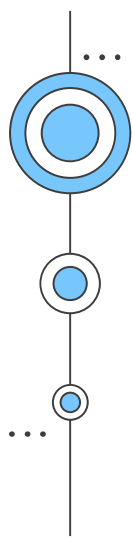
- Can be downloaded to Windows or Linux
- Used within Azure Cloud Shell
- Upload, download, sync or transfer files and blobs

```
azcopy [command] [arguments] --[flag-name]=[flag-value]
```

```
azcopy copy 'file-link-with-sas-key' 'container-link-with-sas-key'
```

| Command                       | Description                                          |
|-------------------------------|------------------------------------------------------|
| <a href="#">azcopy copy</a>   | Copies source data to a destination location         |
| <a href="#">azcopy list</a>   | Lists the entities in a given resource.              |
| <a href="#">azcopy remove</a> | Delete blobs or files from an Azure storage account. |
| <a href="#">azcopy make</a>   | Creates a container or file share.                   |





# Azure Migrate



# Azure Migrate

*Centralized platform that provides guidance and planning for migrations*

## Pre-migration steps



Discover



Dependency analysis



Assess

Cost analysis

Tools to help with migrations



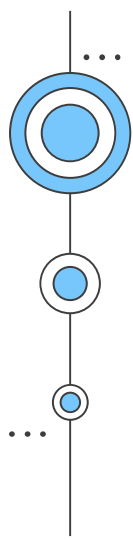
Servers

Databases

Virtual Machines

Web Apps

Data



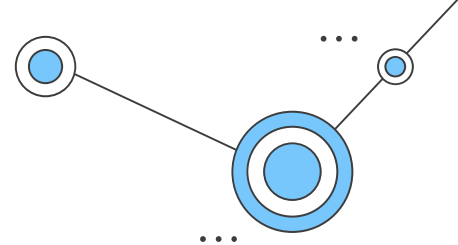
# Data Box



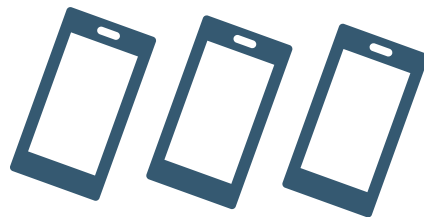


# Data Box

*Device to transfer TBs of data in and out of Azure*



Data upload can very time consuming



Data Box device

Ordered via portal



Regional carrier



Azure datacenter

Quick

Inexpensive

Reliable

Secure

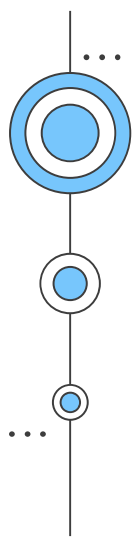
# Data Box

## Use-cases

- Import & export
- > 40 TB
- No to limited network connectivity
- ☐ One-time migration
- ☐ Initial bulk transfer
- ☐ Periodic incremental transfers

## Workflow

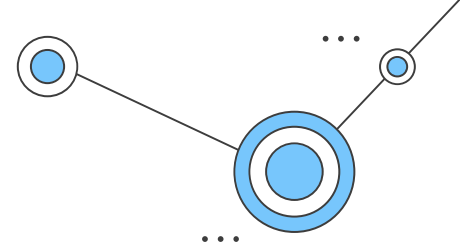




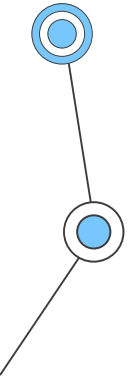
# Azure Marketplace

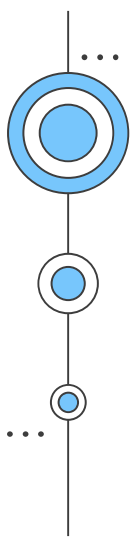


# Azure Marketplace



- **Third-party companies** offer additional applications and services
- Accessed from within Azure portal
- Everything has been certified





# Summary



# Summary

## Storage Account

Cloud solution for storing data  
Account that offeres different storage services

## Redundancy options

Locally redundant storage (LRS)

Zone-redundant storage (ZRS)

Geo-redundant storage (GRS)

Geo-zone-redundant storage (GZRS)

## Access Tier

Hot

Cool

Archive

## Blob Storage

Solution to store massive amounts of unstructured data  
Any type of data: Images, documents, backups, videos

## Queue Storage

Storing large numbers of messages

## Azure Files

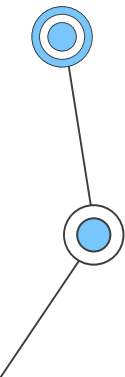
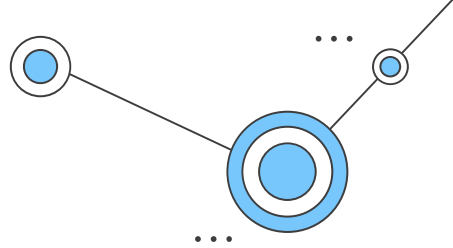
Managed file shares in the cloud  
Can be mounted by cloud or on-premise  
Replace or supplement on-premises file servers

## Azure Sync

Sync data from on-premises to Azure Files

## Azure Tables

Inexpensive NoSQL database service  
Basic structured data



# Summary

## Disk storage

Storage for virtual machines  
Still pay for storage  
Containers (Blob Storage) used

## AzCopy

Command-line tool to copy data to and from storage accounts

## Storage Explorer

Convenient tool to manage storage resources from Desktop

## Azure Migrate

Centralized platform that with tools for planning migrations

## Data Box

Device to transfer TBs of data in and out of Azure

## Azure Marketplace

Trusted third-party companies offer additional applications